BCD & ASCII Conversion

0	0000	5	0101
1	0001	6	0110
2	0010	7	0111
3	0011	8	1000
4	0100	9	1001

Unpacked BCD

- lower 4 bits represents the BCD value

0x04 ⇒ ~~0x00~~ 0100
4

packed BCD

byte — upper nibble (BCD) | lower nibble (BCD)

0x47 ⇒ 0100 0111

Real Time clock ⇒ sec, time, day, month, year

SWAP Rd ; swap nibbles

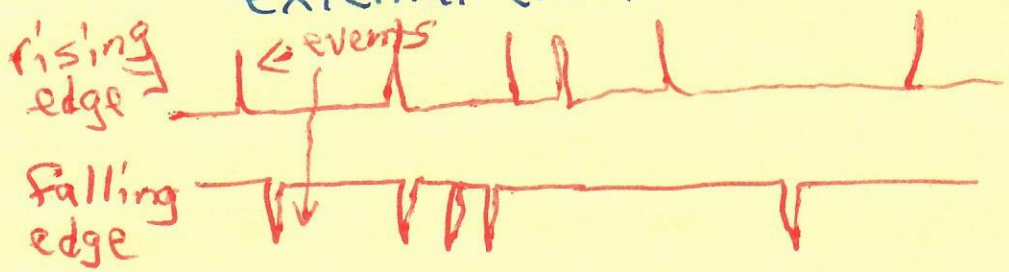
```
LDI R16, 0x48
SWAP R16 ; R16= 0x84
```

Chapter 9 ⇒ timer/counter

Microcontroller applications commonly involve
① detecting/counting events and ② generating periods of time or time delays

① → asynchronous
② → synchronous

events ⇒ requires an external clock source
⇒ counter <u>increments</u> with each external event



periods of time or time delays
⇒ connecting oscillator to clock pin for counter
↳ ⇒ $\frac{1 CC}{1 MC}$ ⇒ AVR ⇒ Atmega32
↳ ⇒ counter <u>increments</u> once every machine cycle (MC) ⇒ increments once every clock cyc

if we want a delay of 50μs
$f_{osc} = 1 MHz \Rightarrow t_{MC} = \left(\frac{1 CC}{1 MC}\right)\left(\frac{1 s}{10^6 CC}\right) = 1 \mu s/MC$
then we need a counter of 50

50 ⇒ find 50 in binary

⇒ 00110010

take 2s comp

11001101

\+ 1

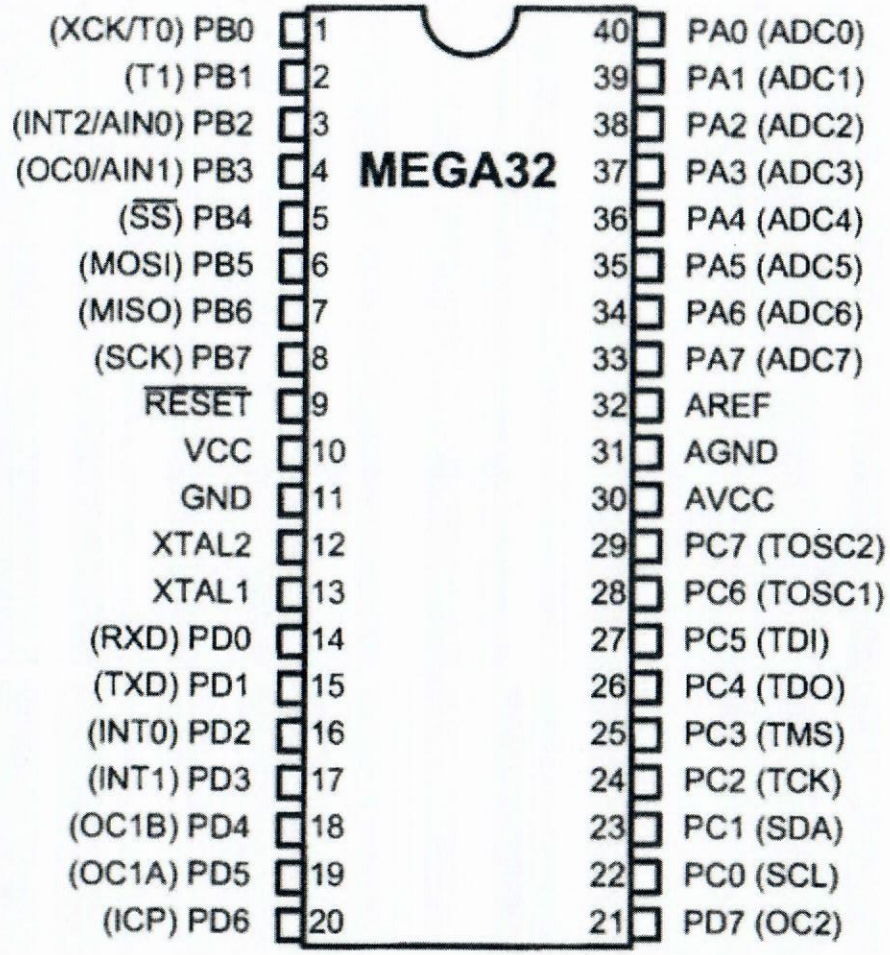
counter ⇒ 1100 1110

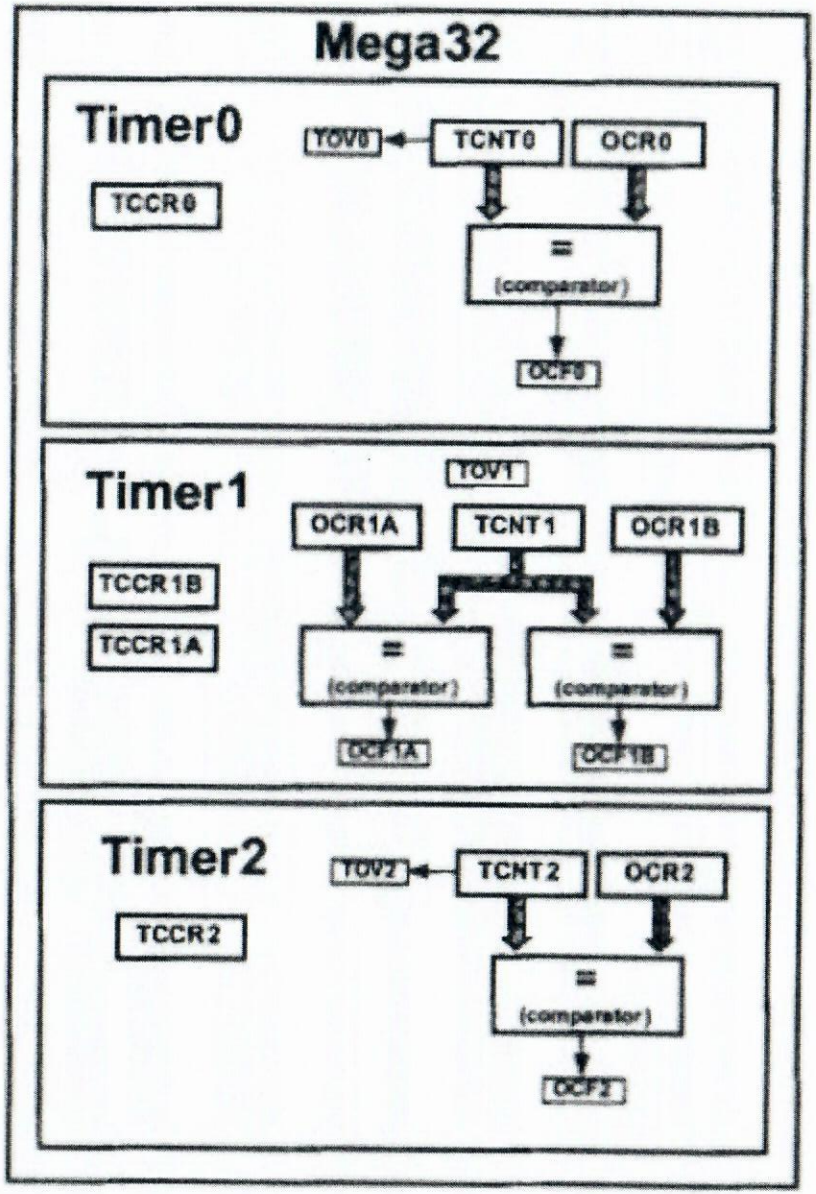
C E

0xCE → 0xCF → 0xD0 → . . .

→ 0xFF → 0x00

Chapter 4-Fig4-1




Mega32
Timer0
TOV0
TCNT0
OCR0
TCCR0
=
(comparator)
OCF0
Timer1
TOV1
OCR1A
TCNT1
OCR1B
TCCR1B
TCCR1A
=
(comparator)
=
(comparator)
OCF1A
OCF1B
Timer2
TOV2
TCNT2
OCR2
TCCR2
=
(comparator)
OCF2

Timer/Counter Registers:

① TCNTn ↑ increments
TCNT0 TCNTn = 0
TCNT1 (a b) load TCNTn
TCNT2

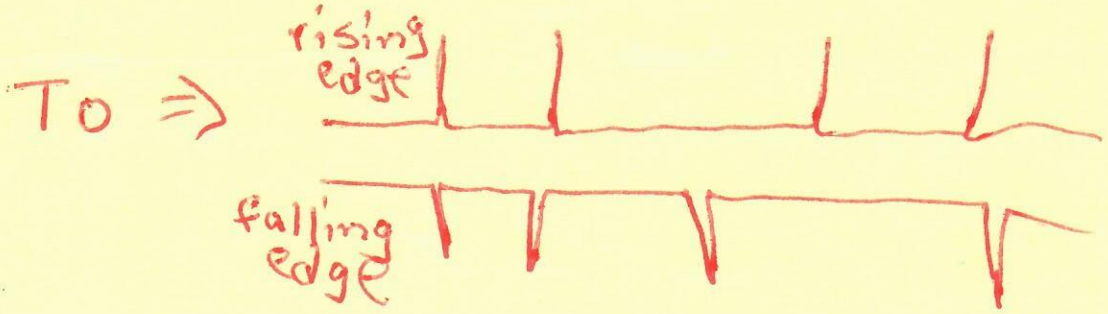
OUT I/O Registers

② TOVn (timer/counter overflow) (TIFR)
TIFR ⇒ timer interrupt flag register

TOV0
TOV1 TOVn = 0 if no overflow
TOV2 TOVn = 1 if overflow

resetting flag write a 1 to TOVn & 0s to other bits of TIFR

③ TCCRn (Timer/counter control Register)



④ OCRn (output Compare register)
OCR0 = 0x23 ⇒ When TCNT0 = 0x23
OCFn (output compare match flag) ⇒ TCNT0 = 0 (clear)
⇒ OCF0 = 1



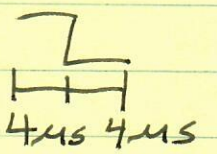
CS02:CS00

001 ⇒ F_1 ⇒ clk (no prescaling)

$$F_8 = \frac{1MHz}{8} \Rightarrow T = \frac{8}{10^6} = 8\mu s$$

CS02:CS00

010



Create a square wave of 50% duty cycle on PB.5 (PortB, bit5). $f_{osc} = 8$ MHz

```
.ORG 0
LDI R16, HIGH(RAMEND)
OUT SPH, R16
LDI R16, LOW(RAMEND)
OUT SPL, R16  ; move SP when using
                subroutines

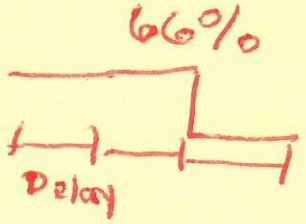
SBI DDRB, 5 ;  PORTB.5 ⇒ output
LDI R16, 0x20 ; 0b 00 1 0000
                   5 4 3 2 1 0
                   2     0

LDI R17, 0
OUT PORTB, R17 ; PORTB = 0x00 ⇒ init PB
Begin: CALL Delay
PB.5   EOR R17, R16 ;
OUT PORTB, R17
RJMP Begin
```

x	y	x⊕y	
0	0	0	
0	1	1 ←	toggle PB.5
1	0	1	
1	1	0 ←	

```
SBI PORTB, 5
call Delay
SBI PORTB, 5
call Delay
CBI PORTB, 5
call Delay
```

⇒



AVR
ICC
IMC

normal
COM01:COM00

Delay: LDI R20, 194;
OUT TCNT0, R20; TCNT0 = 194
[redacted], [redacted]; 00 000001
CLK
WGM01:WGM00
normal
LDI R20, 0x01;
OUT TCCR0, R20;

timer 0
normal mode
internal clock
no prescaler

FOC0 (0) for [redacted] our purposes

TCNT0 → 0xFF → 0x00

Again: IN R20, TIFR; TIFR ⇒ TOV0 flag
SBRS R20, TOV0; if TOV0=1, skip ⇒ timer overflow
RJMP Again;
TOV0=0
LDI R20, 0x00;
OUT TCCR0, R20; turn off timer0 (CS02:CS00 000)
LDI R20, (1<<TOV0); 00000001
OUT TIFR, R20; clear TOV0 flag write 1 to TOV0 0s to other bits
RET

$f_{osc} = 8\,MHz \Rightarrow T_{MC} = \left(\frac{1cc}{1MC}\right)\left(\frac{1s}{8\times10^6\,cc}\right)$

$T_{MC} = 0.125\,\mu s/MC$

$\frac{1cc}{1MC}$

194
preload

194 preload

$$\begin{array}{r} 256 \\ -\,194 \\ \hline 62 \end{array}$$

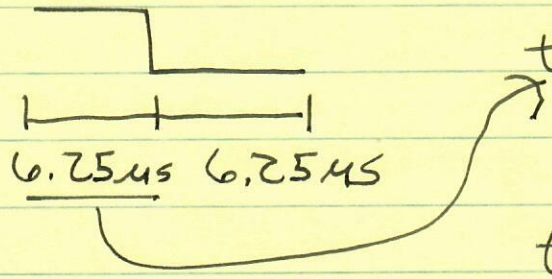
-62
$C2$

194 → hex

LDI R20, -62
OUT TCNT0, R20

Assume $f_{osc} = 8\,MHz$. Write a program to generate an 80kHz square wave on PORTC bit 2 (RC2). Timer 0

$f = 80kHz \Rightarrow T = \frac{1}{f} = \frac{1}{80(10^3)} = 12.5\mu s$



$t_{delay} = n t_{MC}$

$t_{MC} = \left(\frac{1\,cc}{TMc}\right)\left(\frac{1s}{8(10^6)\,cc}\right)$

$= 0.125\mu s/MC$

$n = \frac{6.25\mu s}{0.125\mu s} = 50$ ticks

TCNTO $= -50$ or $256 - 50 = 206$

hex

CE

```
;  in class example - 80 kHz square wave


.ORG 0
LDI R16, HIGH(RAMEND)
OUT SPH, R16
LDI R16, LOW(RAMEND)
OUT SPL, R16; MOVE SP BECAUSE USING SUBROUTINE

SBI DDRC, 2; PC2 OUTPUT
LDI R17, 0X04; 0B00000100 BIT POSITION 2 SET
LDI R18, 0
OUT PORTC, R18; INITIALIZE PC2 = 0

BEGIN:      RCALL DELAY
            EOR R18, R17; TOGGLE PC2
            OUT PORTC, R18; UPDATE PC2
            RJMP BEGIN

            ; TIMER 0 DELAY
DELAY:      LDI R21, -50; PRELOAD FOR NUMBER OF TICKS FOR HALF WAVE
            OUT TCNT0, R21
            LDI R21, 0X01; 0B00000001-LAST 3 BITS 001=> CLK NO PRESCALER
            OUT TCCR0, R21
AGAIN:      IN R21, TIFR
            SBRS R21, TOV0; CHECK STATUS OF TOV0
            RJMP AGAIN; WAIT FOR TIMER0 TO OVERFLOW (TOV0 SET)
            LDI R21, 0X00
            OUT TCCR0, R21; TURN OFF TIMER 0
            LDI R21, (1<<TOV0); RESET TOV0 (WRITE A 1 TO IT, 0S TO OTHER BITS)
            OUT TIFR, R21
            RET
```

```
DELAY MACHINE CYCLES

;  in class example - 80 kHz square wave


.ORG 0
LDI R16, HIGH(RAMEND)
OUT SPH, R16
LDI R16, LOW(RAMEND)
OUT SPL, R16; MOVE SP BECAUSE USING SUBROUTINE

SBI DDRC, 2; PC2 OUTPUT
LDI R17, 0X04; 0B00000100 BIT POSITION 2 SET
LDI R18, 0
OUT PORTC, R18; INITIALIZE PC2 = 0

BEGIN:     RCALL DELAY
           EOR R18, R17; TOGGLE PC2
           OUT PORTC, R18; UPDATE PC2
           RJMP BEGIN

      MCs   ; TIMER 0 DELAY
DELAY: 1    LDI R21, -50; PRELOAD FOR NUMBER OF TICKS FOR HALF WAVE
       1    OUT TCNT0, R21
       1    LDI R21, 0X01; 0B00000001-LAST 3 BITS 001=> CLK NO PRESCALER
       1    OUT TCCR0, R21
AGAIN:  1   IN R21, TIFR
       2/1  SBRS R21, TOV0; CHECK STATUS OF TOV0
       2    RJMP AGAIN; WAIT FOR TIMER0 TO OVERFLOW (TOV0 SET)
       1    LDI R21, 0X00
       1    OUT TCCR0, R21; TURN OFF TIMER 0
       1    LDI R21, (1<<TOV0); RESET TOV0 (WRITE A 1 TO IT, 0S TO OTHER BITS)
       1    OUT TIFR, R21
       4    RET
```

DELAY MACHINE CYCLES => 1 LDI + 1 OUT + 1 LDI + 1 OUT (ONLY EXECUTED ONCE)
+ (AGAIN LOOP) (1 IN + 1 SBRS + 2 RJMP) 13 TIMES + 1 IN + 2 SBRS
+ 1 LDI + 1 OUT+ 1 LDI + 1 OUT + 4 RET (INSTRUCTIONS AFTER AGAIN LOOP)
= 4+[13(4)+3]+8 = 67 MCS (ACTUAL COMPARED TO TARGETED 50 MCS FOR TIMER0)
ERROR = 17/50 = 34%

AGAIN LOOP EXPLANATION

(12 TIMES TOV0 NOT SET, IN 13TH ITERATION TOV0 NOT SET WHEN IN INSTRUCTION EXECUTED, SO MUST COMPLETE 13TH ITERATION) AND THE FIRST 2 INSTRUCTIONS (IN, SBRS FOR 14TH ITERATION SO THAT TOV0 IS DETECTED)TO GET 50 MCS THEN MUST GO THROUGH LOOP AGAIN BECAUSE TOV0 SET BUT NOT DETECTED IN SBRS INSTRUCTION YET SO + 1 IN + 1 SBRS

Bit	7	6	5	4	3	2	1	0
	FOC0	WGM00	COM01	COM00	WGM01	CS02	CS01	CS00
Read/Write	W	RW	RW	RW	RW	RW	RW	RW
Initial Value	0	0	0	0	0	0	0	0

FOC0 D7 Force Output compare: Writing 1 to it forces the wave generator to act as if a compare match has occurred.

WGM01:00	D6	D3	Timer0 mode selector bits
	0	0	Normal
	0	1	CTC (Clear Timer on Compare match)
	1	0	PWM, phase correct
	1	1	Fast PWM

COM01:00 D5 D4 Compare Output Mode; The table shows what the wave generator does on compare match when the timer is in Normal or CTC mode:

COM01	COM00	Description
0	0	Normal port operation, OC0 disconnected
0	1	Toggle OC0 on compare match
1	0	Clear OC0 on compare match
1	1	Set OC0 on compare match

CS02:00 D2D1D0 Timer0 clock selector

0	0	0	No clock source (Timer/Counter stopped)
0	0	1	clk (no prescaling)
0	1	0	clk / 8
0	1	1	clk / 64
1	0	0	clk / 256
1	0	1	clk / 1024
1	1	0	External clock source on T0 pin. Clock on falling edge
1	1	1	External clock source on T0 pin. Clock on rising edge

Figure 15-4. TCCR0 (Timer/Counter Control Register) Register

Bit	7	6	5	4	3	2	1	0
	FOC0	WGM00	COM01	COM00	WGM01	CS02	CS01	CS00
Read/Write	W	RW	RW	RW	RW	RW	RW	RW
Initial Value	0	0	0	0	0	0	0	0

FOC0 D7 Force compare match: This is a write-only bit, which can be used while generating a wave. Writing 1 to it causes the wave generator to act as if a compare match had occurred.

WGM00, WGM01

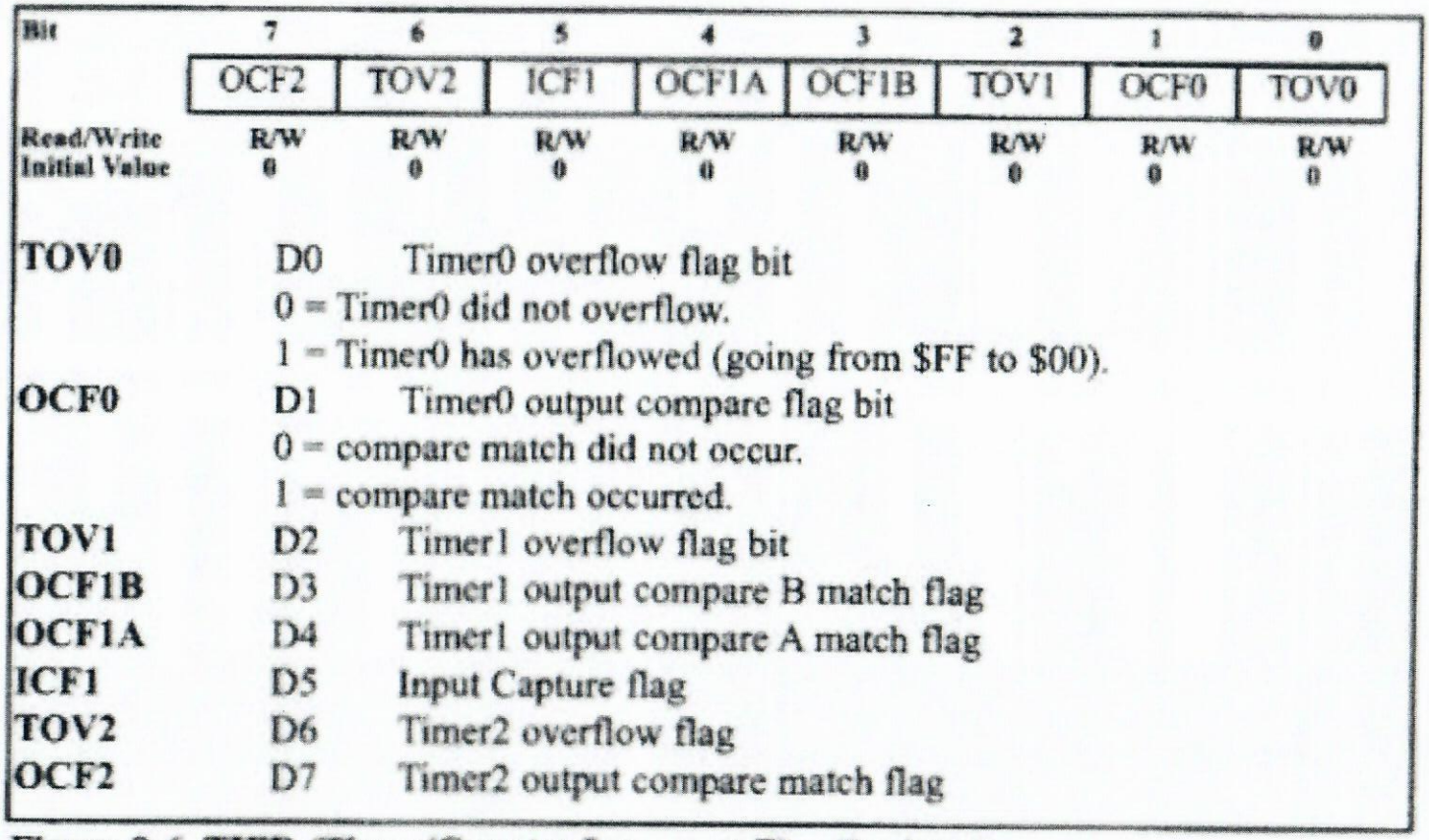
D6	D3	Timer0 mode selector bits
0	0	Normal
0	1	CTC (Clear Timer on Compare Match)
1	0	PWM, phase correct
1	1	Fast PWM

COM01:00 D5 D4 Compare Output Mode:
These bits control the waveform generator (see Chapter 15).

CS02:00

D2	D1	D0	Timer0 clock selector
0	0	0	No clock source (Timer/Counter stopped)
0	0	1	clk (No Prescaling)
0	1	0	clk / 8
0	1	1	clk / 64
1	0	0	clk / 256
1	0	1	clk / 1024
1	1	0	External clock source on T0 pin. Clock on falling edge.
1	1	1	External clock source on T0 pin. Clock on rising edge.

Figure 9-5. TCCR0 (Timer/Counter Control Register) Register



Bit	7	6	5	4	3	2	1	0
	OCF2	TOV2	ICF1	OCF1A	OCF1B	TOV1	OCF0	TOV0
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial Value	0	0	0	0	0	0	0	0

TOV0 D0 Timer0 overflow flag bit
0 = Timer0 did not overflow.
1 = Timer0 has overflowed (going from $FF to $00).

OCF0 D1 Timer0 output compare flag bit
0 = compare match did not occur.
1 = compare match occurred.

TOV1 D2 Timer1 overflow flag bit

OCF1B D3 Timer1 output compare B match flag

OCF1A D4 Timer1 output compare A match flag

ICF1 D5 Input Capture flag

TOV2 D6 Timer2 overflow flag

OCF2 D7 Timer2 output compare match flag

Figure 9-6. TIFR (Timer/Counter Interrupt Flag Register)